

APPROACH TO APPARENTLY MINOR BLUNT HEAD TRAUMA IN THE CHILDREN'S EMERGENCY

BACKGROUND

Blunt minor head trauma is a common presentation in CE across all ages. Most injuries are mild and are not associated with long term neurological sequelae. 5% of apparently minor blunt head trauma patients have traumatic brain injury (TBI) on imaging and 1% with clinically important TBI (ciTBI).

GENERAL CONSIDERATIONS

- Assessment of the pre-verbal child (< 2-years-old) is challenging for the following reasons:
 - Skull fractures or ciTBI may occur despite minor head trauma
 - Intracranial injuries are often asymptomatic
- In infants < 12-months, scalp hematomas are predictive of skull fractures; there is an increased risk of underlying skull fractures with
 - Non-frontal hematomas, and
 - Large (boggy) hematomas > medium (easily palpable) > small (barely palpable) hematomas
- Consider Non-Accidental Injury (NAI) if injuries are not consistent with the mechanism of fall or if history provided is not consistent with child's expected developmental milestones

INVESTIGATIONS

XR skull

- Limited role in the evaluation of minor blunt head trauma
- Possible indications include screening for fractures in asymptomatic toddlers with significant non-frontal scalp hematomas

CT head

- Gold standard in diagnosing acute TBI
- We need to be aware of the radiation risks associated with CT imaging

MANAGEMENT (refer to flow chart below)

Observation in CE

- Duration of 4 – 6H (from time of head injury)
- Hourly vital signs and CLC
- Discharge criteria:
 - Asymptomatic after observation
 - Normal neurological examination
 - Successful trial of feeds
 - Reliable caregiver who understands head injury advice

Indications for admission

- Infants < 3 months
- Abnormal CT findings
- Persistent symptoms despite normal CT head
- Other injuries that require inpatient management
- Suspected NAI

ADMITTING DISCIPLINE

- NEUROSURGERY – admission for observation/monitoring of head injuries
- PAEDS MEDICINE – admission for management of medical issues/problems with incidental head injury

Please discuss with senior on shift regarding admitting discipline if in doubt.

SUPPLEMENTARY INFORMATION

GLASGOW COMA SCALE (GCS) SCORING

- **Glasgow Coma Scale (Age > 2-years)**

Eye opening (E4)	Verbal response (V5)	Motor response (M6)	Score
		Obeys commands	6
	Oriented	Localizes pain	5
Spontaneous	Confused, disoriented	Withdraws	4
To command	Inappropriate words	Abnormal flexion to pain	3
To pain	Incomprehensible sounds	Abnormal extension to pain	2
None	None	None	1

- **Glasgow Coma Scale (Age ≤ 2-years)**

Eye opening (E4)	Verbal response (V5)	Motor response (M6)	Score
		Spontaneous purposeful movements	6
	Age-appropriate vocalization, smile, or orientation to sound, interacts, follow objects	Withdraws to touch	5
Spontaneous	Cries, irritable	Withdraws to pain	4
To sound	Cries to pain	Abnormal flexion to pain	3
To pain	Moans to pain	Abnormal extension to pain	2
None	None	None	1

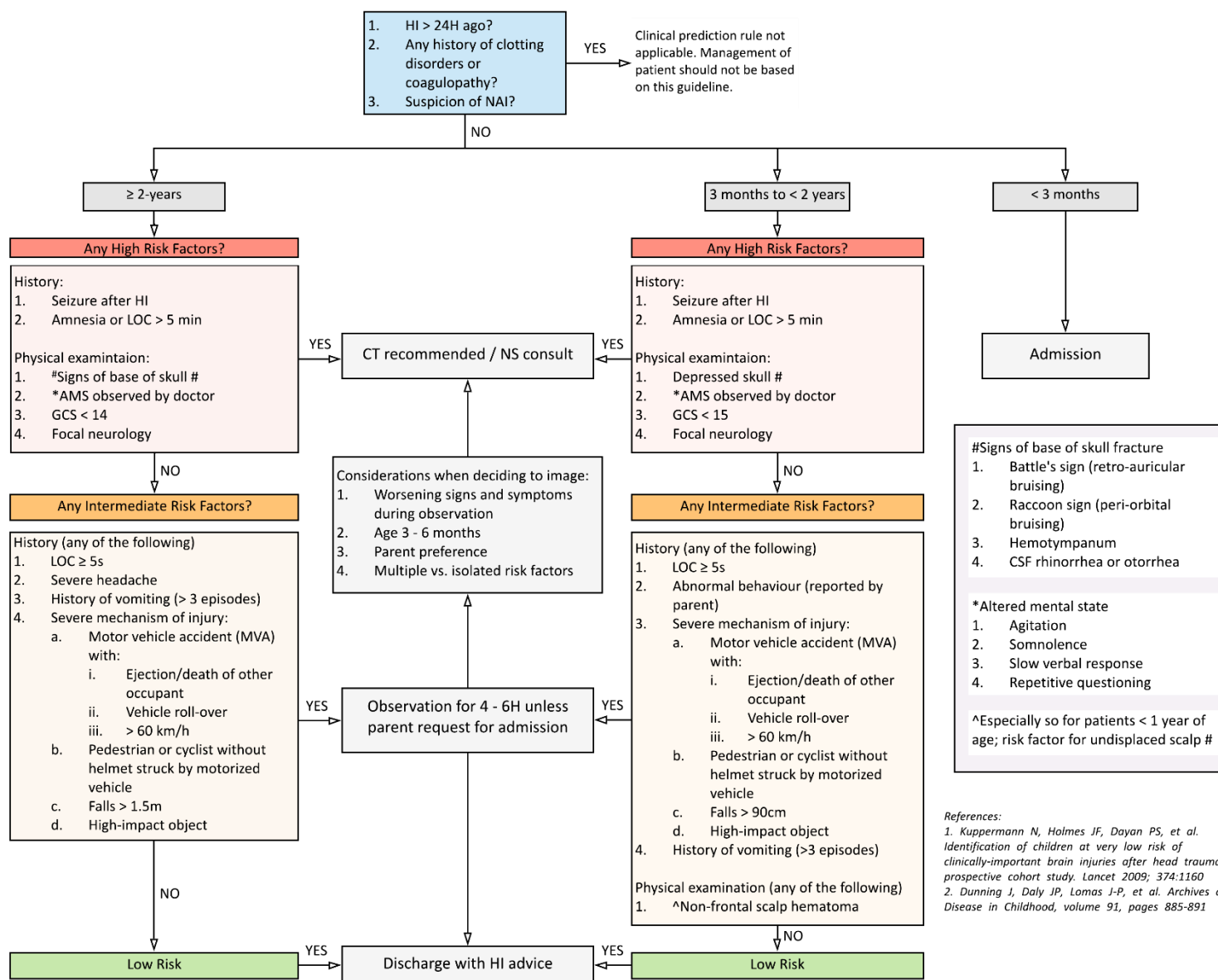
REFERENCES

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3. Greenes DS, Schutzman SA. Clinical significance of scalp abnormalities in asymptomatic head-injured infants. *Pediatr Emerg Care*. 2001;17(2):88-92. doi:10.1097/00006565-200104000-00002

Disclaimer: This document reflects what is currently regarded as safe practice. However, as in any clinical situation, there may be factors which cannot be covered by a single set of guidelines. This document does not replace the need for the application of clinical judgement to each individual presentation.

Revised by Ian Tan Kai Zhi (Version date: July 2020)

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